SYED AFFAN ALI

Mobile Application Developer • Embedded Systems & Robotics • AI / ML Enthusiast

affan.syed001@gmail.com (+92) 3073086688 Karachi, Pakistan linkedin.com/in/syed-affan-dev

PROFESSIONAL SUMMARY

Innovative Computer Systems Engineering graduate with 1.5+ years of professional mobile application development experience across three companies, specializing in Flutter/Dart, embedded systems, and robotics. Demonstrated ability to deliver production-grade mobile applications with complex state management, native API integration, and offline-first architectures. Now actively expanding expertise into Artificial Intelligence, Machine Learning, and Computer Vision — combining a strong hardware-software foundation with modern AI paradigms to build intelligent, real-world systems. Holds IELTS Band 6.5 and is seeking advanced research opportunities in AI, robotics, and intelligent systems.

WORK EXPERIENCE

Mobile Application Developer

Gentec Pvt. Ltd. | Karachi, Pakistan | 11/2024 – Present

- Contributing to production-level in-house mobile application development, collaborating within cross-functional agile teams following industry-standard version control and CI/CD workflows.
- Implementing feature modules using Flutter/Dart with emphasis on clean architecture, scalable state management, and code reusability across platforms.
- Performing structured task delivery within sprint cycles under senior developer mentorship, consistently meeting deadlines in a fast-paced development environment.
- Participating in daily standups, code reviews, and retrospectives — strengthening collaborative engineering practices.

Mobile Developer · IoT & Connected Apps

Wamtsol Pvt. Ltd. | Karachi, Pakistan | 02/2024 – 10/2024

- Developed and maintained cross-platform mobile applications using Flutter, delivering responsive, pixel-perfect UIs aligned with design specifications.
- Implemented advanced state management solutions (Provider, Bloc) enabling scalable, maintainable application architectures for growing feature sets.
- Integrated native device APIs including GeoLocation, Camera, and Media Access — enabling rich, hardware-level feature sets within Flutter apps.
- Utilized local persistence solutions (IsarDB, SharedPreferences) and engineered background data uploads via Dart isolates for reliable offline-first experiences.
- Optimized widget trees and rendering performance, reducing frame drop occurrences and improving app responsiveness on low-end devices.

Flutter Front-End Developer

BaijiSoft Technologies | Karachi, Pakistan | 08/2023 – 01/2024

- Translated Figma UI/UX designs into high-fidelity Flutter interfaces, applying performance-oriented widget composition and rendering best practices.
- Implemented REST API communication using HTTP GET, POST, and multipart requests, with structured JSON data modeling for seamless frontend-backend integration.
- Contributed to design system discussions, advocating for consistent UI patterns and accessibility-friendly component architectures.
- Gained cross-device adaptability experience, ensuring UI consistency across diverse screen sizes and Android/iOS platforms.

Mobile Application Developer Intern

Zensoft Pvt. Ltd. | Karachi, Pakistan | 06/2023 – 07/2023

- Built a production Sugar Cane Trolley Registration App using Flutter with native GeoLocation and Image Picker APIs, deployed for real field operations.
- Implemented nested navigation, structured state management, and integrated Google Maps SDK for location-aware features.

- Engineered offline-first background data upload pipeline using Dart isolates, ensuring data integrity without active connectivity.
- Applied unit testing, async programming, and app lifecycle management practices — delivering a robust, error-resilient application.

EDUCATION

Bachelor of Engineering — Computer Systems

Mehran University of Engineering & Technology (MUET) | Jamshoro, Pakistan | 2019 – 2023

CGPA: 2.88 (3.34 GPA in Final Year)

TECHNICAL SKILLS

Mobile Development	Flutter · Dart · Provider · Bloc · IsarDB · SharedPreferences · Dart Isolates · REST APIs · JSON · HTTP/Multipart
AI / ML	Python · TensorFlow · PyTorch · Scikit-learn · Hugging Face Transformers · LangChain · RAG Pipelines · Prompt Engineering
Computer Vision	OpenCV · Image Processing · Object Detection · Gesture Recognition · Real-time Video Analysis
Agentic AI	LangChain Agents · Tool Use · LLM Orchestration · OpenAI API · Autonomous Pipelines
Embedded & Robotics	Arduino | ESP32 | C / C++ | Microcontroller Interfacing | Sensor Integration | Wireless Telemetry | Gesture-Controlled Robotics | IoT
Languages	Dart · Python · C · C++ · JavaScript (basic)
Tools & Practices	Git | GitHub | Figma | VS Code | Agile/Scrum | Unit Testing | SOLID Principles | Clean Architecture
Languages (Spoken)	Urdu (Native) · English — IELTS Band 6.5 (Proficient)

PROJECTS

2D Ultrasonic Radar Mapping Prototype (Poor Man's LiDAR) | *Arduino | HC-SR04 | Stepper Motor / Servo | Python | Matplotlib | USB Serial*

- Built a low-cost 2D mapping prototype that rotates an HC-SR04 ultrasonic sensor with a stepper motor or servo to scan the surrounding environment.
- Streamed angle-and-distance readings from Arduino over USB serial and rendered them on a live Python polar plot using Matplotlib.

IoT-Based Water Quality Monitoring System | *Arduino | Python | pH, Turbidity & Temperature Sensors | IoT | Data Visualization*

- Designed a real-time water-quality monitoring system using Arduino with pH, turbidity, and temperature sensors.
- Built a Python pipeline for normalization, remote visualization, anomaly detection, and threshold-based alerts, with wireless data transmission for remote monitoring.

Gesture-Controlled Smart Robotic Wheelchair | *ESP | Arduino | Gyroscope | Sonar | Proximity & Touch Sensors | GSM | Motor Control*

- Engineered an assistive mobility prototype controlled through hand gestures using ESP and Arduino boards with responsive motor control.
- Integrated gyroscope, sonar, proximity, and touch sensors with GSM connectivity and battery-management circuits for connected operation and safety monitoring.

AI-Powered Document Q&A Agent | *Python | LangChain | OpenAI API | FAISS | RAG | Streamlit*

- Built a retrieval-augmented document assistant that ingests PDF and text files, performs FAISS vector search, and answers questions with source citations.
- Implemented conversational memory and a Streamlit upload-and-query interface for coherent multi-turn document exploration.

AI-Based IoT Energy Monitoring & Anomaly Detection | *ESP32 | Python | Machine Learning | IoT | Sensors*

- Built an ESP32 prototype that streams electrical measurements for remote monitoring and analysis.
- Applied a lightweight anomaly-detection model to flag unusual energy-consumption patterns and present results through a monitoring interface.

Smart Greenhouse Monitoring & Prediction System | *ESP32 | Python | Temperature, Humidity & Light Sensors | Machine Learning | IoT*

- Collected temperature, humidity, and light readings with an ESP32 and made the environmental data available for remote monitoring.
- Applied a simple machine-learning model for near-term condition prediction and actionable threshold alerts.

RESEARCH INTERESTS

AI-Driven Robotics & Autonomous Systems • Computer Vision for Real-World Applications • Multimodal AI & Human-Robot Interaction • Edge AI & Embedded Machine Learning (TinyML) • Large Language Models & Agentic AI • IoT & Intelligent Sensing Systems

CERTIFICATIONS & TRAINING

Mobile App Development with Flutter | MUET | 07/2022 – 12/2022

Arduino & Robotics | Creativo, Karachi | 04/2021 – 06/2021

IELTS English Proficiency | Band Score: 6.5

RESEARCH OBJECTIVE

To pursue a Master's degree in Artificial Intelligence, Robotics, or Intelligent systems—bridging a strong foundation in mobile application development and embedded hardware with cutting-edge AI research. Committed to contributing to innovative solutions at the intersection of computer vision, autonomous robotics, and edge AI, with the goal of producing impactful, publishable research and building next-generation intelligent systems.